Topic 8 — Microbiology, Environment and Applied Science

Covers syllabus (click to expand)

Microbiology: Major Groups (Bacteria, Viruses, Fungi, Protozoa) | Discovery of Bacteria (Antony van Leeuwenhoek), Compound Microscopy (1500–2000x), Bacterial Nutrition (Bacteriochlorophyll vs Chemosynthesis), Gram Staining (Gram-Positive vs Gram-Negative Cell Wall, Peptidoglycan & Teichoic Acids), *Mycoplasma* (Smallest Autonomous Organism, Lacks Cell Wall, Penicillin Resistant) | Synthetic Medium Culturing (Bacteria/Fungi) vs Obligate Host Cells (Viruses) | Reproduction & Spore Formation (Bacteria, Fungi, Algae, Mosses, Ferns) | Acellular Entities: Viruses (DNA/RNA, Capsid), Bacteriophages, Viroids & Prions | Fungi: Chitin Wall, Glycogen Storage, Achlorophyllous, Yeast, Moulds, Dimorphic Fungi, Mushrooms (Edible Wild *Gucchi* / *Morchella* vs Lethal Poisonous Death Cap *Amanita phalloides*) | Symbiosis & Interspecific Interactions: Mutualism (+/+ : Lichens, Mycorrhiza, Fig & Fig Wasp Co-evolution, Corals + *Symbiodinium*), Commensalism (+/0 : Barnacles on Whales, Remora on Sharks, Cattle Egrets, Epiphytes; Inquilinism, Metabiosis, Phoresy), Parasitism (+/- : *Herpes simplex* Endoparasitism), Hyperparasitism (Parasite of Parasite), Amensalism (-/0) | Mycorrhiza (Phosphorus/Nitrogen Uptake & Root Disease Protection) | Lichens (SO_2 Air Pollution Indicators & Pioneers of Succession on Bare Rocks) | Ecosystem Energetics: Producers (Trophic Level-I / Intermediaries between Biotic and Abiotic), Primary Productivity ($g/m^2/\text{year}$), Microbial Decomposition | Oceanic Primary Producers: Diatoms (~45% Marine Organic Production) & Cyanobacteria vs Primary Consumers (Copepods) | Biological Unity of Human Species (*Homo sapiens*) | Microbes in Human Welfare: Dairy (*Lactobacillus*, LAB), Fermented Foods, Alcoholic Fermentation (*Saccharomyces cerevisiae* / Baker's & Brewer's Yeast) | Industrial Organic Acids (*Aspergillus niger*, *Acetobacter aceti*, *Clostridium butyricum*) & Enzymes (Lipases, Pectinases, Streptokinase "Clot Buster") | Bioactive Molecules: Cyclosporin A (*Trichoderma polysporum*) & Statins (*Monascus purpureus*) | Antibiotics: Penicillin (Alexander Fleming, Florey & Chain), Streptomycin (Selman Waksman), Broad vs Narrow Spectrum | Sewage Treatment Plant (STP): Primary vs Secondary Biological Treatment (Flocs, Aerobic Aeration, Activated Sludge, Anaerobic Digestion), BOD & COD Parameters | Biogas / Gobar Gas Plants (Methanogens, IARI & KVIC, CH_4 50–70%) | Drinking Water Purification: Alum Coagulation, Sand Filtration, Chlorination, Coliform / MPN Count | Waste Management: Biodegradable (Wool, Paper, Dung) vs Non-Biodegradable (Plastics, Glass, Aluminium, DDT), Biomagnification vs Bioaccumulation | Bioremediation: "Superbug" (*Pseudomonas putida* / A. M. Chakrabarty) & Oil Spills | Biomedical Waste Color Coding (Yellow, Red, White, Blue) | Major Indian Research Institutions: Lucknow Cluster (CDRI, CIMAP, NBRI, IITR, BSIP, IISR), UP State (IVRI Bareilly, IIVR Varanasi, IGFRI Jhansi, CIRG Mathura), National Apex Centers (ICAR, ICMR, CSIR, IISc, CCMB, NIV Pune, NDRI Karnal, CFTRI Mysuru, NEERI Nagpur) & International Agencies (WHO, FAO, UNEP) | Scientific Pioneers: Hooke, Leeuwenhoek, Jenner, Pasteur, Koch, Lister, Ehrlich, Ross, Salk, Sabin & Indian Legends (J. C. Bose, P. C. Ray, C. V. Raman, Birbal Sahni, M. S. Swaminathan, Verghese Kurien, H. G. Khorana) | Six Indian Seasons (*Shad Ritus*: Vasanta, Grishma, Varsha, Sharad, Hemanta, Shishira) & Agricultural Meteorology | Important National & International Science/Health Days

Sources baked in: NCERT Class 8, 9, 11 & 12 (Microorganisms: Friend and Foe; Why Do We Fall Ill?; Microbes in Human Welfare; Environmental Issues), Lucent General Science (Microbiology, Ecology & Applied Botany/Zoology), Ghatnachakra General Science (Subdivisions, Evolution of Life & Taxonomy, Notes + 208 Question Bank complete integration), UPPCS Prelims PYQs (1990–2025), UKPCS Prelims, BPSC, RAS, MPPCS, CGPCS, and IAS Prelims **Weight:** ★★★ High — heavily tested across UPPCS: microbe–disease pairs, industrial fermentation products (lactic acid, penicillin, streptokinase, statins), biodegradable vs non-biodegradable waste (wool, DDT), BOD parameters, Lucknow/UP scientific institutes (CDRI, IVRI), scientific pioneers (Jenner, Fleming, Raman), and the six Indian seasons (*Shad Ritus*) **Last verified:** September 2026

Current Affairs

Year / Date	Event / Discovery	Must-Score Fact
28 February	**National Science Day** observed nationwide across India.	Celebrated to commemorate the landmark discovery of the **Raman Effect (inelastic scattering of light)** by **Sir C. V. Raman on 28 February 1928** (awarded Nobel Prize in Physics in 1930). It does **not** mark Raman's birthday, Fleming's penicillin, or World Environment Day.
26 November	**National Milk Day** observed annually in India.	Commemorates the birth anniversary of **Dr. Verghese Kurien**, the "Father of the White Revolution" in India and mastermind of **Operation Flood** (launched in 1970 by NDDB, Anand). Coincides with Constitution Day (*Samvidhan Diwas*).
Still Current	**WHO Global Smallpox Eradication (1980).**	Smallpox remains the **only human infectious disease officially certified as completely eradicated globally from nature** through mass worldwide vaccination initiated by **Edward Jenner's** cowpox inoculation (1796).

Year / Date	Event / Discovery	Must-Score Fact
Still Current	**Antimicrobial Resistance (AMR) & "Superbugs"**.	Indiscriminate and incomplete courses of antibiotics induce selective pressure, promoting plasmid-mediated horizontal gene transfer (via conjugation) and bacterial multidrug resistance (e.g., MRSA — *Methicillin-Resistant Staphylococcus aureus*, NDM-1 — *New Delhi Metallo-beta-lactamase-1*).

8.1 The Microbial World: Structure, Staining & Taxonomy

Microbiology is the branch of biological science dealing with microscopic organisms invisible to the naked human eye (< 0.1 mm).

Discovery of Bacteria & Compound Microscopy

- **Antony van Leeuwenhoek (1632–1723)**: Born in Delft, Netherlands; regarded universally as the **Father of Microbiology**; first to discover and accurately describe bacteria and free-living protozoa in 1674, christening them **"animalcules"**.
- **Microscopy & Bacterial Scale**:
- Most bacteria measure between **0.5 to 5.0 micrometres (μm)** in size (though giant sulfur bacteria like *Thiomargarita* can reach 0.5 mm).
- Bacteria cannot be visualized with a simple hand lens; they are observed under a **Compound Microscope**, which provides magnifications of **1,500 to 2,000 times**.

Bacterial Nutrition: Autotrophic vs Heterotrophic

- **Heterotrophic Bacteria**: The overwhelming majority of bacteria are heterotrophs:
- *Saprobes*: Decompose and absorb nutrients from dead organic matter and corpses.
- *Parasites*: Live on or inside living plant and animal hosts, causing infectious diseases (e.g., cholera, tuberculosis, syphilis).
- **Autotrophic Bacteria**: Synthesize organic compounds from inorganic raw materials (CO_2, H_2O):
- *Photosynthetic Autotrophs*: Use solar radiant energy; **they DO NOT have chloroplasts**; instead, they possess specialized photosynthetic pigments called **Bacteriochlorophylls** located in internal membrane folds (thylakoid lamellae).
- *Chemosynthetic Autotrophs*: Synthesize organic food by oxidizing inorganic chemical compounds (e.g., *Nitrosomonas* oxidizes ammonia to nitrite; *Thiobacillus* oxidizes hydrogen sulfide H_2S; methanogens oxidize CO_2 and H_2 into methane).

Bacteria: Morphology & Gram Staining

- **Structure**: Prokaryotic unicellular organisms lacking an organized nucleus, nucleolus, and membrane-bound organelles (mitochondria, chloroplasts, endoplasmic reticulum, Golgi body).
- **Genetic Material**: Consists of a single circular double-stranded DNA molecule lying naked in the cytoplasm in an area termed the **Nucleoid**; extra-chromosomal, self-replicating circular DNA rings called **Plasmids** carry non-essential adaptive genes (e.g., antibiotic resistance and fertility factors).
- **Cell Wall**: Constructed of **Peptidoglycan (Murein)**, a polysaccharide polymer cross-linked by short amino acid peptides.
- **Classification by Shape**: 1. *Coccus* (spherical): *Streptococcus pneumoniae*, *Staphylococcus*. 2. *Bacillus* (rod-shaped): *Bacillus anthracis*, *Lactobacillus*, *Escherichia coli*. 3. *Spirillum* (spiral/corkscrew): *Spirillum*, *Treponema pallidum* (causes Syphilis). 4. *Vibrio* (comma-shaped): *Vibrio cholerae* (causes Cholera).
- **Gram Staining (Christian Gram, 1884)**:
- Microscopic diagnostic technique based on differential cell wall thickness and chemical composition.
- Bacteria are stained with **Crystal Violet**, treated with Gram's **Iodine** (mordant), decolorized with **Ethanol/Acetone**, and counterstained with **Safranin**.
- **Gram-Positive**: Single thick peptidoglycan wall (20–80 nm) containing **Teichoic Acids**; retains the crystal violet-iodine complex; appears **Purple/Violet** under light microscopy (e.g., *Staphylococcus*, *Streptococcus*, *Bacillus*, *Clostridium*).
- **Gram-Negative**: Thin peptidoglycan layer (8–10 nm) covered by an outer lipid-rich membrane composed of **Lipopolysaccharides (LPS)**; alcohol dissolves outer lipid layer, eluting crystal violet; cells take up safranin counterstain, appearing **Pink/Red** (e.g., *E. coli*, *Salmonella*, *Vibrio cholerae*, *Pseudomonas*); outer membrane acts as a protective barrier against penicillin and lysozyme.

> **Note:** *Bacterial Endospores* are not reproductive units; they are extremely resistant survival structures formed under adverse environmental conditions by genera like *Bacillus* and *Clostridium*. High resistance to heat and radiation is mediated by **Calcium**

> **dipicolinate** in the spore cortex.

Mycoplasma: The Smallest Autonomous Cells

- **Mycoplasmas (PPLO — Pleuropneumonia-Like Organisms)**:
- The **smallest known living organisms capable of autonomous growth and reproduction** ($0.1–0.3\ \mu m$).
- Characterized by the **complete lack of a cell wall**, enclosed only by a triple-layered flexible lipoprotein plasma membrane.
- *Antibiotic Fact*: Because they lack a peptidoglycan cell wall, **mycoplasmas are naturally resistant to Penicillin and Beta-Lactam antibiotics** (which function by blocking bacterial cell wall synthesis).
- Many mycoplasmas are pathogenic; *Mycoplasma pneumoniae* causes primary atypical walking pneumonia in humans.

Synthetic Medium Culturing vs Viral Host Cell Obligation

- **Bacteria and Fungi**: Possess independent cellular and metabolic machinery (ribosomes, tRNA, ATP generation); can be readily cultured and grown on **artificial/synthetic nutrient media** (such as nutrient agar, potato dextrose agar, or nutrient broth), forming macroscopic colonies visible to the naked human eye.
- **Viruses**: Are acellular obligate intracellular molecular parasites lacking ribosomes, metabolic enzymes, and ATP synthesis; **CANNOT be grown or cultured in artificial/synthetic cell-free nutrient media**; require living host cells (*in vivo* inside intact animals/embryos or *in vitro* inside living cell cultures) for genomic replication and assembly.

Reproduction & Spore Formation

- Unicellular organisms reproduce primarily by asexual processes (binary fission in bacteria and *Amoeba*, budding in yeast).
- **Spore Formation**: A widespread mode of asexual reproduction where specialized resistant or dispersive reproductive units (spores) develop and germinate into new individuals without gametic fusion.
- *Must-Score*: Spore formation occurs across **Bacteria, Fungi, Algae, and lower non-flowering plants like Mosses (Bryophytes) and Ferns (Pteridophytes)**.

Archaebacteria vs Eubacteria

- Archaebacteria represent the most primitive surviving prokaryotes; differ from true bacteria (*Eubacteria*) by having a cell wall lacking peptidoglycan (composed of pseudomurein) and possessing unique **branched-chain ether-linked membrane lipids** providing resistance to extreme temperatures, salinity, and pH: 1. **Methanogens**: Strictly anaerobic; inhabit swamp mud and cattle rumen; reduce CO_2 or formic acid into **Methane (CH_4)** (*Methanobacterium*). 2. **Halophiles**: Inhabit saturated brines and salt lakes (*Halobacterium*). 3. **Thermoacidophiles**: Inhabit scalding acidic hot sulfur springs at temperatures up to $100°C$ (*Thermoplasma*, *Sulfolobus*).

Viruses: Acellular Obligate Parasites

- **Nature**: Acellular nucleoprotein packages; completely non-living outside a host cell (lack cytoplasm, ribosomes, ATP-generating enzymes, and metabolic machinery); active and multiplying only upon invading a susceptible host cell.
- **Structure**: Core contains either **DNA or RNA** (never both together), surrounded by an outer protective protein coat called the **Capsid**, formed of small sub-units called **Capsomeres**.
- *Plant Viruses*: Mostly contain single-stranded RNA (ssRNA, e.g., TMV).
- *Animal Viruses*: Contain ssRNA (Influenza, Rabies, Polio, HIV, Coronaviruses), dsRNA (Rotavirus), or dsDNA (Smallpox, Herpes, Hepatitis B).
- *Bacteriophages*: Viruses infecting bacteria; predominantly contain double-stranded DNA (dsDNA).
- **Sub-Viral Pathogens**:
- **Viroids**: Discovered by T. O. Diener (1971); free, low-molecular-weight single-stranded circular RNA without a protein capsid; infect plants (Potato Spindle Tuber Viroid).
- **Prions**: Discovered by Stanley Prusiner (Nobel 1997); infectious abnormally folded proteins devoid of nucleic acids; cause fatal spongiform encephalopathies (Mad Cow Disease / BSE in cattle, Scrapie in sheep, CJD in humans).

8.2 Microbes in Human Welfare & Applied Biotechnology

Microorganisms play an indispensable role in food processing, industrial chemical fermentation, sewage recycling, and bio-pharmaceutical therapeutics.

Dairy, Food & Alcoholic Fermentation

- **Curd Production**: Milk is inoculated with a small starter culture (inoculum) containing millions of **Lactic Acid Bacteria (LAB / *Lactobacillus*)**; at ambient temperatures ($37–40^{\circ}C$), LAB ferments milk sugar lactose into lactic acid:

$$\text{Lactose} \xrightarrow{\text{LAB}} \text{Lactic Acid}$$

  Lactic acid drops pH, precipitating and partially digesting the milk protein **Casein**, converting liquid milk into a thick, custard-like curd. LAB substantially improves curd's nutritional value by synthesizing **Vitamin B_{12}** and suppresses harmful putrefying bacteria in the human stomach.

UKPCS 2025

When milk is converted into curd, the sour taste is due to

A. Acetic acid
B. Citric acid
C. Lactic acid
D. Tartaric acid

Show answer

Logic: *Lactobacillus* ferments lactose into lactic acid, imparting the characteristic sour taste to curd. Acetic acid is in vinegar, citric acid is in citrus fruits, and tartaric acid is in tamarind and grapes. **Ans: C.** Lactic acid.

- **Fermentation in Baking & Brewing**:
- **Baker's Yeast (*Saccharomyces cerevisiae*)**: Added to wheat dough; carries out anaerobic fermentation converting glucose into ethanol and carbon dioxide:

$$C_6H_{12}O_6 \xrightarrow{\text{Zymase enzyme}} 2C_2H_5OH + 2CO_2 \uparrow$$

  The escaping bubbles of **Carbon Dioxide (CO_2)** cause the dough to swell and rise (leavening), leaving spongy air pockets when baked into bread.
- **Brewer's Yeast**: Strains of *S. cerevisiae* ferment malted cereals and fruit juices to produce alcoholic beverages:
  - *Non-distilled beverages* (lower alcohol content, 3–12%): **Beer** (from fermented barley) and **Wine** (from fermented grapes).
  - *Distilled beverages* (higher alcohol content, 40–50%): **Whisky** (distilled cereal mash), **Brandy** (distilled wine), **Rum** (distilled sugarcane molasses), and **Vodka**.

Industrial Microbial Metabolites & Wonder Enzymes

- **Organic Acids (NCERT Class 12 Master Pairs)**:
- **Citric Acid**: Industrial acidulant produced by the fungus ***Aspergillus niger***.
- **Acetic Acid (Vinegar)**: Produced by the bacterium ***Acetobacter aceti***.
- **Butyric Acid**: Produced by the anaerobic bacterium ***Clostridium butyricum***.
- **Lactic Acid**: Produced by the bacterium ***Lactobacillus***.
- **Bioactive Enzymes & Medicines**:
- **Lipases**: Produced by fungi like *Candida*; used in laundry detergent powders to break down hydrophobic triglycerides, removing greasy collars and food oil stains.
- **Pectinases & Proteases**: Commercial enzymes added to clarify turbid fruit juices; by degrading insoluble structural pectins and proteins, bottled store-bought juices become sparklingly transparent.
- **Streptokinase**: An enzyme isolated from cultures of the bacterium ***Streptococcus*** and modified through recombinant DNA technology; administered intravenously as a life-saving **"clot buster"** to rapidly hydrolyze intravascular fibrin clots in emergency myocardial infarction patients.
- **Cyclosporin A**: An 11-amino acid cyclic peptide isolated from the soil fungus ***Trichoderma polysporum***; acts as a targeted **immunosuppressant** that selectively inhibits T-cell activation, routinely prescribed to prevent organ transplant rejection.
- **Statins**: Fermentation products isolated from the red yeast ***Monascus purpureus***; competitively binds to the active site of the enzyme **HMG-CoA reductase**, blocking the rate-limiting step of cholesterol synthesis in the liver, functioning as a commercial blood cholesterol-lowering medication (e.g., Lovastatin).

Antibiotics & Landmark Milestones

- **Discovery of Penicillin (1928)**:
- While working on *Staphylococcus aureus* bacteria at St. Mary's Hospital, London, Scottish bacteriologist **Sir Alexander Fleming** noticed a clear zone of lysis around a contaminating greenish mould colony in an unwashed culture plate.
- The mould was identified as ***Penicillium notatum*** (an ascomycete fungus), and the bactericidal substance was named **Penicillin**.
- A decade later, Oxford researchers **Howard Florey and Ernst Chain** developed industrial extraction and purification methods, establishing penicillin as a systemic life-saving therapeutic during World War II. Fleming, Florey, and Chain shared the 1945 Nobel Prize.

UKPCS 2025

Penicillin was discovered by

A. E. J. Butler
B. Waldeyer
C. Alexander Fleming
D. Robert Hooke

Show answer

Logic: Alexander Fleming observed in 1928 that the mould *Penicillium notatum* produced an antibacterial compound that killed *Staphylococcus* cultures. Robert Hooke discovered and named the cell (1665), Waldeyer named chromosomes (1888), and E. J. Butler is recognized as the Father of Indian Plant Pathology. **Ans: C.** Alexander Fleming.

- **Streptomycin**: Isolated from the soil actinomycete bacterium ***Streptomyces griseus*** in 1943 by **Selman Waksman** and his student Albert Schatz; represented the first effective clinical cure for Tuberculosis (*Mycobacterium tuberculosis*); Waksman coined the term "Antibiotic" and won the 1952 Nobel Prize.
- **Spectrum of Antibiotics**:
- *Narrow-spectrum*: Active predominantly against either Gram-positive or Gram-negative bacteria (e.g., natural Penicillin G).
- *Broad-spectrum*: Active across a wide range of both Gram-positive and Gram-negative pathogenic organisms (e.g., Tetracyclines, Chloramphenicol, Ampicillin).

8.3 Important Microbial Diseases & Historical Traps

Bacterial vs Viral Diseases (The Essential Distinctions)

Disease	Causative Pathogen	Pathogen Class	Mode of Transmission	Major Clinical Features & Diagnostic Tests
Cholera	*Vibrio cholerae*	Comma bacterium	Contaminated drinking water / food	Profuse "rice-water" stools, severe dehydration, hypovolemic shock.
Typhoid	*Salmonella typhi*	Rod bacterium	Faecal–oral route / contaminated food	Sustained step-ladder fever, intestinal perforation; **Widal Test**.
Tuberculosis	*Mycobacterium tuberculosis*	Acid-fast rod bacterium	Airborne droplet nuclei from cough	Chronic cough with hemoptysis, night sweats, cavitation; **Mantoux Test**, DOTS.
Tetanus	*Clostridium tetani*	Spore-forming rod bacterium	Contaminated deep puncture wounds	Neurotoxin causes rigid spasms of jaw (**Lockjaw**), arching of spine.
Whooping Cough	*Bordetella pertussis*	Coccobacillus bacterium	Airborne droplets	Paroxysmal coughing ending in deep inspiratory "whoop"; prevented by **DPT**.
Hib Meningitis	*Haemophilus influenzae* type b	Coccobacillus bacterium	Droplets / respiratory secretions	Severe pediatric meningitis, epiglottitis, pneumonia; prevented by **Pentavalent Vaccine**.
Diphtheria	*Corynebacterium diphtheriae*	Rod bacterium	Respiratory droplets	Leathery pseudomembrane over tonsils/pharynx suffocating airway; **Schick Test**, DPT.
Syphilis	*Treponema pallidum*	Spirochaete bacterium	Sexual contact / congenital	Hard painless chancre ulcer, secondary skin rash, tertiary neurosyphilis; **VDRL Test**.

Disease	Causative Pathogen	Pathogen Class	Mode of Transmission	Major Clinical Features & Diagnostic Tests
Gonorrhoea	*Neisseria gonorrhoeae*	Diplococcus bacterium	Sexual contact	Purulent urethral discharge, burning micturition, pelvic inflammation.
Malaria	*Plasmodium vivax / falciparum*	Sporozoan Protozoan	Bite of infective female *Anopheles*	Chills and recurring high fever (haemozoin toxin release), anemia, splenomegaly.
Amoebiasis	*Entamoeba histolytica*	Amoeboid Protozoan	Contaminated food/water (cysts)	Amoebic dysentery with blood and mucus in stool, abdominal cramps, liver abscess.
Kala-Azar	*Leishmania donovani*	Flagellated Protozoan	Bite of Sandfly (*Phlebotomus*)	Visceral leishmaniasis, hepatosplenomegaly, severe anemia, skin darkening.
Sleeping Sickness	*Trypanosoma gambiense*	Flagellated Protozoan	Bite of Tsetse fly (*Glossina*)	African trypanosomiasis, lymphadenopathy, day-time sleepiness, coma.
Ringworm	*Trichophyton / Microsporum*	Filamentous Fungus	Direct skin contact / infected towels	Circular scaly intensely itchy erythematous skin lesions with clear centers.

UPPCS (Pre) 2022

The bacillus *Haemophilus influenzae* causes—

A. A form of meningitis in young children
B. Influenza
C. Whooping cough
D. Pneumonia

Show answer

Logic: *Haemophilus influenzae* type b is a premier bacterial pathogen causing acute purulent **meningitis** and severe pneumonia in infants and young children under 5 years of age. It does **not** cause Influenza (which is caused by the Influenza virus). Whooping cough is caused by *Bordetella pertussis*. In the 2022 official key sheet, an erroneous provisional key was initially published due to confusion with the obsolete synonym *Haemophilus pertussis*; the scientifically correct primary disease is meningitis in young children. **Ans: A.** A form of meningitis in young children.

Great Scientific Pioneers & Inventions

- **Antonie van Leeuwenhoek (1674)**: "Father of Microbiology"; handcrafted high-magnification single-lens microscopes; first human to observe and describe living single-celled organisms, naming them **"animalcules"** (bacteria, protozoa, sperm, and red blood corpuscles).
- **Edward Jenner (1796)**: "Father of Immunology"; recognized that English milkmaids infected with bovine cowpox never contracted deadly smallpox; took fluid from cowpox pustules on milkmaid Sarah Nelmes and inoculated young James Phipps, conferring complete immunity against smallpox; created the world's **first vaccine** (*vacca* is Latin for cow).
- **Louis Pasteur (1860s–1880s)**: "Father of Modern Microbiology"; completely shattered the doctrine of Spontaneous Generation (*Abiogenesis*) using goose-neck flasks; demonstrated that fermentation is driven by living yeast cells; developed **Pasteurisation** (heating liquids to $63^{\circ}C$ for 30 minutes or $71.7^{\circ}C$ for 15 seconds followed by rapid chilling to $< 10^{\circ}C$ to kill non-spore-forming spoilage pathogens); developed life-saving vaccines for **Rabies** (Hydrophobia), Anthrax, and Fowl Cholera.
- **Robert Koch (1876–1882)**: "Father of Medical Bacteriology"; developed methods of culturing bacteria on solid agar media; isolated the bacterial pathogens causing **Anthrax** (*Bacillus anthracis*), **Tuberculosis** (***Mycobacterium tuberculosis***, announced on 24 March 1882), and **Cholera** (***Vibrio cholerae***); formulated **Koch's Postulates**: 1. The specific microorganism must be consistently present in diseased individuals and absent in healthy individuals. 2. The organism must be isolated from the host and grown in pure culture outside the body. 3. The pure culture must reproduce the specific disease when inoculated into a healthy, susceptible animal. 4. The same microorganism must be recovered again from the experimentally infected host.
- **Joseph Lister (1865)**: "Father of Modern Antiseptic Surgery"; applied Pasteur's microbial germ theory to hospital surgery by spraying volatile **Carbolic Acid (Phenol)** onto surgical wounds, incisions, dressings, and instruments, cutting postoperative gangrene and sepsis mortality from 50% to under 15%.
- **Paul Ehrlich (1910)**: "Father of Chemotherapy"; introduced the concept of the **"Magic Bullet"** (a chemical that selectively attacks pathogenic microbes without injuring host tissues); synthesized **Salvarsan (Arsphenamine / Compound 606)** to cure Syphilis.
- **Jonas Salk (1954) & Albert Sabin (1961)**: Salk developed the injected **Inactivated Polio Vaccine (IPV)** using formaldehyde-killed viruses; Sabin developed the live-attenuated **Oral Polio Vaccine (OPV)** administered orally as liquid drops.

8.4 Ecosystem Energetics, Interspecific Interactions & Waste Management

Ecosystem Structure & Energetics

- **Biotic vs Abiotic Components**:
- **Abiotic Factors**: Non-living physical and chemical elements — solar irradiance, temperature, water, atmospheric gases, and soil minerals.
- **Biotic Community**: Living organisms interacting within an ecological habitat.
- **Producers as Intermediaries**: Autotrophic green plants, photosynthetic algae, and cyanobacteria absorb solar radiant energy and convert inorganic molecules (CO_2, H_2O) into organic carbohydrates; thus, **Producers act as the fundamental intermediary between the abiotic physical environment and the biotic community**.
- **Trophic Hierarchy**:
  - **Trophic Level-I**: Autotrophs / Green plants (Primary Producers).
  - **Trophic Level-II**: Herbivores (Primary Consumers).
  - **Trophic Level-III**: Carnivores feeding on herbivores (Secondary Consumers).
  - **Trophic Level-IV**: Apex predators / Top carnivores (Tertiary Consumers).
- **Primary Productivity**:
- The rate at which solar radiant energy is captured and stored as chemical biomass by autotrophic green plants through photosynthesis per unit area over a specified time interval.
- Expressed quantitatively as $\mathbf{g/m^2/year}$ (biomass) or $\mathbf{kcal/m^2/year}$ (energy).
- **Microbial Decomposition**:
- The biological breakdown of complex organic matter (leaf litter, dead animal carcasses, faecal wastes) into simple inorganic nutrients ($CO_2, H_2O, NO_3^-, PO_4^{3-}$) by saprotrophic microorganisms (**bacteria and fungi**).
- **Primary Producers in Oceanic Food Chains**:
- Marine ecosystems cover ~71% of Earth's surface and contribute nearly half of global net primary production.
- **Diatoms**: Microscopic unicellular algae with ornate silica cell walls (frustules); **contribute ~45% of total oceanic primary organic production**.
- **Cyanobacteria**: Photosynthetic blue-green prokaryotes (e.g., *Prochlorococcus*, *Synechococcus*) performing oxygenic photosynthesis.
- *Common Traps*:
  - **Copepods**: Microscopic crustaceans that are **primary consumers / secondary producers** (feed on diatoms).
  - **Foraminifera**: Heterotrophic amoeboid protists possessing calcium carbonate shells (*tests*); they consume bacteria and micro-detritus, acting as consumers.

Interspecific Biological Interactions & Symbiosis

Organisms in natural ecosystems engage in constant interspecific ecological relationships: 1. **Mutualism ($+/+$)**: Reciprocally beneficial interaction where both species gain: - **Lichens**: Mutualistic association between an alga/cyanobacterium (**Phycobiont**) and a fungus (**Mycobiont**). The fungus absorbs water, minerals, and provides anchorage, while the photosynthetic partner provides synthesized carbohydrates. - *Pioneers of Succession*: Lichens secrete organic acids that erode bare rocks into soil, initiating primary xerarch/lithosere succession. - *Bio-indicators of Pollution*: Lichens have no cuticle or roots; they absorb airborne toxins directly; **exceptionally sensitive to Sulfur Dioxide ($\mathbf{SO_2}$)**; they **do NOT grow in polluted industrial/urban areas**. - **Mycorrhiza**: Mutualism between soil fungi and roots of higher vascular plants (gymnosperms and angiosperms). - *Benefits to Plant*: Massively increases the root surface area for **absorption of Phosphorus, Nitrogen, and trace minerals**; confers drought tolerance; and provides **protection against root-borne fungal and bacterial pathogens**. - **Fig and Fig Wasp (*Blastophaga): A classic example of tight obligate co-evolutionary mutualism. A specific fig tree species can only be pollinated by its specific partner wasp species; the female wasp lays eggs inside the fig fruit inflorescence and pollinates the flowers, while the developing seeds provide food for the wasp larvae. - Coral Reef Symbiosis*: Scleractinian corals (Cnidarians) live in obligate endosymbiosis with intracellular photosynthetic dinoflagellates of the genus** Symbiodinium ***(zooxanthellae). The dinoflagellates nourish the coral polyps in nutrient-poor tropical waters. - *Root Nodule Bacteria**: *Rhizobium* forms a mutualistic symbiotic association in the root nodules of leguminous crops (peas, beans, clover), fixing gaseous N_2 into plant-usable ammonium compounds (NH_4^+). 2. **Commensalism ($+/0$)**: An interaction where one organism (the commensal) benefits, while the host organism is neither harmed nor benefited: - **Barnacles on Whales**: Sessile marine crustaceans attach to the skin of baleen whales to gain free locomotive transport to nutrient-rich waters (*Phoresy*). - **Remora (Suckerfish) & Sharks**: Remora attaches to sharks using a modified dorsal fin sucker,

feeding on discarded scraps of the shark's prey without affecting the shark. - **Cattle Egrets & Grazing Cattle**: Egrets forage in close proximity to cattle, catching insects disturbed and flushed out from grass by the cattle's footsteps. - **Epiphytes (Orchids)**: Plants that grow physically on the branches of large canopy trees solely for access to sunlight, absorbing moisture from rain and air without penetrating the host tissue or robbing nutrients. 3. **Parasitism (+/−)**: One organism (parasite) benefits by living on or inside the host, deriving nutrition and metabolic energy while causing physiological harm: - **Endoparasitism**: Parasites living internally inside host organs/cells; e.g., the ***Herpes simplex* virus** replicating inside human epithelial and nerve cells, depending entirely on host cellular machinery. - **Ectoparasitism**: Parasites feeding externally on the surface of host bodies; e.g., female mosquitoes, ticks, mites, and human head lice sucking host blood. 4. **Hyperparasitism**: - The phenomenon where a **parasite parasitizes another parasite** (a parasite within a parasite). - Especially prevalent among parasitoid wasps in the suborder **Apocrita** (Hymenoptera), certain flies (**Diptera**), and beetles (**Coleoptera**). 5. **Species Concept & Human Unity**: - A biological species is a reproductive community whose members can interbreed and produce fertile offspring. - All modern human populations — including Chinese, American, Indian, and Black African ethnic groups — belong to the **exact same biological species**: ***Homo sapiens***.

Biodegradable vs Non-Biodegradable Pollutants

- **Classification of Waste**:
- **Biodegradable Waste**: Organic substances capable of being decomposed naturally by microbial enzymes within weeks or months; e.g., domestic food waste, animal manure, human faeces/urine, paper, vegetable scraps, and natural animal/plant fibres:
  - **Wool**: Composed of the structural protein **Keratin**; completely digested by soil proteolytic bacteria and fungal dermatophytes.
  - **Cotton & Jute**: Composed of the polysaccharide **Cellulose**; degraded by cellulolytic microorganisms (*Trichoderma*, *Cellulomonas*).
- **Non-Biodegradable Waste**: Synthetic, mineral, or recalcitrant substances that resist microbial enzymatic hydrolysis and remain virtually unchanged in nature for centuries:
  - **Plastics**: High-molecular-weight synthetic polymers (polyethylene, polypropylene, polyvinyl chloride / PVC, polystyrene); lack natural enzymatic recognition sites; break physically into microplastics (< 5 mm) that accumulate in marine and terrestrial food webs.
  - **Metals & Glass**: Aluminium foil, tin cans, lead, mercury, and silicate glass containers; completely non-biodegradable.
  - **Synthetic Chemical Pesticides**: **DDT** (Dichloro-diphenyl-trichloroethane), **BHC / Lindane**, and **Endosulfan**; persistent synthetic organochlorines.

UPPCS (Pre) 2022

Which of the following materials is biodegradable?

A. Plastic cups
B. Wool
C. Glass bottle
D. Aluminium foil

Show answer

Logic: Standard key matches Wool. **Ans: B.** Wool

UPPCS (Pre) 2022

Dichloro-diphenyl-trichloroethane (DDT) is a—

A. Non-pollutant
B. Non-biodegradable pollutant
C. Biodegradable pollutant
D. Biochemical pollutant

Show answer

Logic: DDT is a synthetic organochlorine pesticide resistant to microbial breakdown. It persists in soil and water for decades, dissolves in animal fat, and undergoes biomagnification across food chains. Hence, it is a non-biodegradable pollutant. **Ans: B.** Non-biodegradable pollutant.

UPPCS (Pre) 2023

Which of the following is **not** a biodegradable pollutant?

A. Domestic waste
B. Urine
C. Faecal matter
D. Pesticide

Show answer

Logic: Domestic organic kitchen waste, urine, and faecal matter are biological wastes rapidly degraded into basic nutrients by saprophytic bacteria and fungi. Chemical pesticides are synthetic compounds resistant to microbial breakdown, classifying them as non-biodegradable pollutants. **Ans: D.** Pesticide.

> **Note:** *Recyclable vs Biodegradable*: Glass and aluminium are 100% recyclable because they can be melted, purified, and reshaped without degradation of properties. However, they are **0% biodegradable**; microbes cannot metabolize them into soil humus.

Biomagnification & Environmental Toxicology

- **Biomagnification (Biological Magnification)**:
- Phenomenon wherein non-biodegradable, lipid-soluble toxicants that cannot be metabolized or excreted by organisms undergo **progressive exponential increases in concentration at successively higher trophic levels** of a food chain.
- Classic demonstration: **DDT in an Aquatic Food Chain**:

$$\text{Water (0.003 ppb)} \rightarrow \text{Zooplankton (0.04 ppm)} \rightarrow \text{Small Fish (0.5 ppm)} \rightarrow \text{Large Fish (2 ppm)} \rightarrow \text{Fish-eating Birds (25 ppm)}$$

- Toxicological consequence: In apex predatory birds (e.g., pelicans, ospreys, bald eagles, peregrine falcons), high DDT concentrations inhibit the enzyme **Carbonic Anhydrase** in the shell gland, disrupting calcium deposition. This results in **abnormally thin, fragile eggshells** that crack under the weight of brooding parents, causing catastrophic reproductive failure and species decline.
- **Bioremediation**:
- The technological use of living microorganisms to remove, degrade, or detoxify environmental contaminants from contaminated soil and groundwater:
  - **"Superbug" (*Pseudomonas putida*)**: Developed in 1971 by Indian-American geneticist **Dr. Ananda Mohan Chakrabarty**; by combining four plasmids (CAM, OCT, XYL, NAH) via conjugation into a single bacterium, this strain efficiently digests complex multi-ring aromatic hydrocarbons, octane, and xylene in crude oil, enabling rapid clean-up of **oceanic oil spills**; in 1980, the US Supreme Court (*Diamond v. Chakrabarty*) granted it the world's first patent for a genetically modified organism.
  - **Lichens**: Dual organisms highly sensitive to atmospheric **Sulfur Dioxide (SO_2)**; serve as natural bio-indicators of clean air (absent near industrial complexes and highway junctions).

Sewage Treatment Plants (STP) & Water Quality Parameters

- **Sewage Treatment Workflow**: 1. **Primary Treatment (Physical Separation)**: Raw domestic sewage is passed through mesh wire bar screens to filter out floating physical debris (rags, paper, plastic bottles). The sewage enters a primary settling tank where heavy inorganic grit, sand, and pebbles settle out by gravity, forming **Primary Sludge**. The floating supernatant is the **Primary Effluent**. 2. **Secondary Treatment (Biological Oxidation)**:
  - Primary effluent is transferred into massive, continuously churned **Aeration Tanks**, where air is vigorously bubbled through the liquid.
  - This stimulates explosive growth of aerobic heterotrophic microorganisms forming **Flocs** (dense clumps of bacteria entwined with fungal mycelial filaments).
  - These flocs actively consume the bulk of dissolved organic pollutants, dramatically lowering the **Biochemical Oxygen Demand (BOD)** of the water. 3. **Settling & Anaerobic Sludge Digestion**:
  - Once BOD is substantially reduced, the effluent is passed into a secondary settling tank where flocs settle down by gravity as **Activated Sludge**.
  - A small fraction of activated sludge is pumped back into the aeration tank to act as an **inoculum**.
  - The remaining bulk of sludge is transferred into sealed **Anaerobic Sludge Digesters**.
  - Here, anaerobic methanogenic bacteria digest the aerobic bacteria and fungi, liberating **Biogas** — a combustible mixture of **Methane (CH_4)**, **Carbon Dioxide (CO_2)**, and traces of Hydrogen Sulfide (H_2S). 4. **Tertiary Treatment (Physicochemical**

Disinfection): Before final discharge into natural rivers, the treated water is disinfected to kill remaining pathogenic microbes using **Chlorine gas**, **Bleaching powder**, **Ozone (O_3)**, or **Ultraviolet (UV) radiation**.

BOD vs COD: The Key Analytical Metrics

- **Biochemical Oxygen Demand (BOD)**:
- Definition: The amount of dissolved oxygen in milligrams (mg/L or ppm) required by aerobic microorganisms to biologically oxidize all the biodegradable organic matter present in 1 litre of a water sample incubated at $20°C$ for 5 days (BOD_5).
- Significance: **Direct quantitative measure of organic pollution**. Higher organic contamination → higher microbial activity → higher BOD → severe depletion of **Dissolved Oxygen (DO)** in water bodies.
- Scale:
  - Pristine drinking water: $BOD < 1\ \text{mg/L}$.
  - Moderately polluted river: $BOD = 2\text{–}5\ \text{mg/L}$.
  - Severely polluted sewage: $BOD = 100\text{–}400\ \text{mg/L}$.
  - **Fish Mortality Rule**: Normal dissolved oxygen in clean surface water is $8\text{–}10\ \text{mg/L}$. If sewage discharge drives DO below $4\ \text{mg/L}$, fish and aquatic invertebrates suffocate and die en masse.
- **Chemical Oxygen Demand (COD)**:
- Definition: The amount of oxygen required to chemically oxidize all organic compounds (both biodegradable and recalcitrant/non-biodegradable synthetic chemicals) in a water sample using a powerful boiling chemical oxidizer, **acidified Potassium Dichromate ($K_2Cr_2O_7$)**.
- *Must-Score*: In any polluted water sample, **COD is always significantly higher than BOD** ($\text{COD} > \text{BOD}$), because COD measures the chemical oxidation of *all* organic matter, whereas BOD measures only biologically degradable organics.

Drinking Water Purification & Potability Standards

- **Municipal Water Treatment Steps**: 1. *Coagulation / Flocculation*: Addition of **Potash Alum** ($K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$); the trivalent Al^{3+} ions neutralize negative surface charges on microscopic colloidal clay particles, causing them to aggregate into dense, rapidly settling flocs. 2. *Rapid Sand Filtration*: Passing water through stratified beds of anthracite, sand, and gravel to filter out fine particulate suspensions. 3. *Disinfection (Chlorination)*: Dosing with Chlorine gas (Cl_2) or Bleaching Powder ($CaOCl_2$). Chlorine reacts with water to generate **Hypochlorous Acid ($HOCl$)**, which penetrates microbial cell walls and oxidizes critical metabolic enzymes, killing cholera, typhoid, and dysentery pathogens.
- **Coliform Count as an Indicator Organism**:
- It is technically impossible to test municipal water for every individual viral and bacterial pathogen. Instead, water quality laboratories test for **Faecal Coliform Bacteria** (primarily ***Escherichia coli***).
- *Rationale*: *E. coli* is a normal, non-pathogenic inhabitant of the mammalian intestinal tract. Its presence in a water sample confirms direct contamination with human sewage or animal faeces.
- Standard for Potable Water: **Zero coliforms per 100 mL of drinking water** (tested via Membrane Filtration or Most Probable Number / MPN technique).

Biogas / Gobar Gas Plant Architecture

- Pioneered in India by the **Indian Agricultural Research Institute (IARI)** and the **Khadi and Village Industries Commission (KVIC)**.
- Consists of a cylindrical concrete digester pit (10–15 feet deep) fitted with a floating steel or fixed concrete gas holder.
- Raw cattle dung (*gobar*) is rich in cellulose from ingested grass and naturally hosts symbiotic **Methanogenic Archaebacteria** (***Methanobacterium***) derived from the bovine rumen.
- Cattle dung is blended with water in a 1:1 ratio to form a slurry, fed into the digester, and sealed anaerobically.
- Anaerobic fermentation proceeds in three phases: 1. *Hydrolysis*: Complex carbohydrates and proteins broken into soluble monomers. 2. *Acidogenesis*: Volatile organic acids (acetic, propionic) produced. 3. *Methanogenesis*: Methanogens convert acetic acid, CO_2, and H_2 into **Methane Gas**.
- **Biogas Composition**:
- **Methane (CH_4)**: **50–70%** (primary combustible component; high calorific value $\approx 20\ \text{MJ/m}^3$).
- **Carbon Dioxide (CO_2)**: **30–40%**.
- **Trace Gases**: Hydrogen (H_2), Nitrogen (N_2), and Hydrogen Sulfide (H_2S, 1–2%).
- Advantages: Non-polluting, smokeless fuel for rural cooking and lighting; spent effluent slurry is an exceptional odourless bio-fertiliser rich in nitrogen, phosphorus, and potassium (NPK).

8.5 Major Indian Scientific Institutions, Pioneers & The Six Seasons

Premier Research Institutions of India (High-Yield Match-Pairs)

UP hub	Institutes
Lucknow cluster	**CDRI** (drugs); **CIMAP** (medicinal & aromatic plants); **NBRI** (botany); **IITR** (toxicology); **BSIP** (palaeosciences); **IISR** (sugarcane)
Bareilly (Izatnagar)	**IVRI** (veterinary); **CARI** (avian)
Varanasi	**IIVR** (vegetable research)
Jhansi	**IGFRI** (grassland & fodder)
Mathura	**CIRG** (goats)
Kanpur	**NSI** (National Sugar Institute)
Saharanpur	**CPPRI** (pulp & paper)

National Apex Centers & Specialized Laboratories

- **ICAR (Indian Council of Agricultural Research)**: New Delhi — apex national coordinating body for agricultural research and education.
- **IARI (Indian Agricultural Research Institute / "Pusa Institute")**: New Delhi — premier institute for crop breeding, soil science, and agro-technology.
- **ICMR (Indian Council of Medical Research)**: New Delhi — apex biomedical research and funding council in India.
- **CSIR (Council of Scientific and Industrial Research)**: New Delhi — established in 1942; operates 37 premier national laboratories; its first Director-General was Dr. Shanti Swaroop Bhatnagar.
- **IISc (Indian Institute of Science)**: Bengaluru — founded in 1909 with the visionary philanthropy of Jamsetji N. Tata.
- **TIFR (Tata Institute of Fundamental Research)**: Mumbai — established in 1945 by Dr. Homi J. Bhabha.
- **CCMB (Centre for Cellular and Molecular Biology)**: Hyderabad — flagship institute for genomics, proteomics, and cellular diagnostics.
- **CDFD (Centre for DNA Fingerprinting and Diagnostics)**: Hyderabad — apex forensic and genetic diagnostic center.
- **NIV (National Institute of Virology)**: Pune — apex Indian laboratory for tracking emerging viral pathogens (COVID-19, Nipah, Zika, Chandipura).
- **NDRI (National Dairy Research Institute)**: Karnal (Haryana) — pioneer in dairy processing, cattle cross-breeding, and buffalo somatic cloning.
- **CFTRI (Central Food Technological Research Institute)**: Mysuru (Karnataka) — food processing, preservation, and nutritional chemistry.
- **NEERI (National Environmental Engineering Research Institute)**: Nagpur (Maharashtra) — environmental monitoring, water quality, and EIA assessments.
- **CPRI (Central Potato Research Institute)**: Kufri, Shimla (Himachal Pradesh) — potato seed breeding and disease research.
- **NRRI (National Rice Research Institute, formerly CRRI)**: Cuttack (Odisha) — rice breeding and germplasm conservation.
- **CAZRI (Central Arid Zone Research Institute)**: Jodhpur (Rajasthan) — desertification control, arid crop development.

Apex International Agencies

- **WHO (World Health Organization)**: Headquarters in **Geneva, Switzerland** (founded 7 April 1948).
- **FAO (Food and Agriculture Organization)**: Headquarters in **Rome, Italy** (founded 16 October 1945).
- **UNEP (United Nations Environment Programme)**: Headquarters in **Nairobi, Kenya** (founded 1972).
- **WMO (World Meteorological Organization)**: Headquarters in **Geneva, Switzerland**.
- **IRRI (International Rice Research Institute)**: Headquarters in **Los Baños, Philippines** (developed *IR-8* "Miracle Rice").
- **CIMMYT (International Maize and Wheat Improvement Center)**: Headquarters in **Texcoco, Mexico** (birthplace of semi-dwarf Green Revolution wheat developed by Dr. Norman Borlaug).

Indian Scientific Legends & Inventions

- **Acharya Jagadish Chandra Bose (1858–1937)**:
- Pioneer of wireless radio communication (generated millimetre-wavelength microwaves before Marconi).
- Designed the **Crescograph** (capable of magnifying plant movement and cellular expansion up to 10,000 times).
- Proved experimentally that plants possess nervous irritability, cellular pulsatile contractions, electrical conduction analogous to animals, and undergo physical fatigue when stimulated repeatedly.
- **Acharya Prafulla Chandra Ray (1861–1944)**:
- Recognized as the "Father of Indian Chemistry".
- Synthesized stable **Mercurous Nitrite** in 1896.
- Founded **Bengal Chemical and Pharmaceutical Works Ltd.** (1901) in Kolkata, initiating India's indigenous modern pharmaceutical manufacturing industry.
- **Sir Chandrasekhara Venkata Raman (C. V. Raman) (1888–1970)**:
- Discovered that when monochromatic light passes through a transparent liquid or gas, a tiny fraction of the scattered light emerges with altered wavelengths (frequencies) characteristic of the molecular bonds of the material (**Raman Effect**).
- Discovery announced on **28 February 1928**; awarded the **Nobel Prize in Physics in 1930** (first Asian to receive a science Nobel).
- In commemoration, the Government of India designated **28 February** as **National Science Day**.
- **Prof. Birbal Sahni (1891–1949)**:
- "Father of Indian Palaeobotany".
- Pioneered the study of Indian fossil floras from the Rajmahal volcanic traps of Bihar/Jharkhand; erected the new extinct gymnosperm group ***Pentoxyleae*** (including *Pentoxylon sahnii*).
- Founded the Institute of Palaeobotany at Lucknow (now named Birbal Sahni Institute of Palaeosciences / BSIP).
- **Prof. M. S. Swaminathan (1925–2023)**:
- "Father of the Green Revolution in India".
- Agronomist and plant geneticist who collaborated with Nobel laureate Dr. Norman Borlaug to introduce semi-dwarf, fertilizer-responsive Mexican wheat germplasm (*Sonora 64*, *Lerma Rojo*) into India, breeding iconic rust-resistant high-yielding varieties (*Kalyan Sona*, *Sonalika*).
- Transformed India from a "ship-to-mouth" famine-threatened importer into a self-sufficient grain exporter; recipient of the first World Food Prize (1987) and posthumously honored with the Bharat Ratna (2024).
- **Dr. Verghese Kurien (1921–2012)**:
- "Father of the White Revolution" / "Milkman of India".
- Engineered **Operation Flood** (launched in 1970 by the National Dairy Development Board), the world's largest agricultural dairy development program.
- Pioneered the industrial manufacture of milk powder and condensed milk from buffalo milk (previously considered technically unfeasible) at the Kaira District Cooperative Milk Producers' Union (**Amul**), Anand, Gujarat; elevated India into the world's largest milk producer.
- **Dr. Har Gobind Khorana (1922–1999)**:
- Indian-American biochemist who shared the 1968 Nobel Prize in Physiology or Medicine with Marshall Nirenberg and Robert Holley.
- Synthesized defined oligoribonucleotides to conclusively decipher the **triplet genetic code**, demonstrating which nucleotide codons direct the incorporation of specific amino acids into proteins; subsequently synthesized the world's first fully artificial functional gene in 1970.

The Six Traditional Indian Seasons (*Shad Ritus*)

The traditional Vedic and classical calendar divides the astronomical solar year into six distinct seasons (*Ritus*), each spanning approximately two lunar months:

Order	Season (*Ritu*)	Approx. window
1	**Vasanta** (Spring)	Mid-Feb – Mid-Apr
2	**Grishma** (Summer)	Mid-Apr – Mid-Jun
3	**Varsha** (Monsoon)	Mid-Jun – Mid-Aug
4	**Sharad** (Autumn)	Mid-Aug – Mid-Oct

Order	Season (*Ritu*)	Approx. window
5	**Hemanta** (Pre-winter)	Mid-Oct – Mid-Dec
6	**Shishira** (Winter)	Mid-Dec – Mid-Feb

1. **Vasanta (Spring — वसन्त)**: - *Period*: Chaitra – Vaishakha (approx. mid-February to mid-April). - *Climate*: Moderate pleasant temperatures, departure of severe winter cold, gentle southern breezes (*Malayavanila*). - *Nature*: Deciduous trees shed old foliage and produce vibrant tender leaves; mango trees burst into bloom (*Manjari*); peak activity of pollinators, honeybees, and butterflies; celebration of Holi and Chaitra Navratri. - *Agriculture*: Maturation and harvesting of winter **Rabi crops** (mustard, gram, barley, early wheat).
2. **Grishma (Summer — ग्रीष्म)**: - *Period*: Jyeshtha – Ashadha (approx. mid-April to mid-June). - *Climate*: Severe solar radiation, dry heat, desiccating winds (**Loo** across northern plains), drying of ponds and lowering of water tables. - *Nature*: Forest fires, physiological water stress in wild animals and cattle; flowering of Gulmohar and Amaltas (*Cassia fistula*). - *Agriculture*: Harvesting of late wheat; cultivation of fast-maturing irrigated **Zaid crops** (Watermelon, Muskmelon, Cucumber, Bitter gourd, Moong) in river basin sandbars.
3. **Varsha (Monsoon / Rainy — वर्षा)**: - *Period*: Shravana – Bhadrapada (approx. mid-June to mid-August). - *Climate*: Ingress of the South-West Monsoon; heavy convective rainfall, overcast skies, high ambient relative humidity. - *Nature*: Transformation of parched brown landscapes into lush green; emergence of earthworms and amphibians; explosive surge in vector mosquitoes (*Anopheles*, *Aedes*, *Culex*), triggering seasonal spikes in Malaria, Dengue, Chikungunya, and water-borne Cholera/Gastroenteritis. - *Agriculture*: Sowing of principal **Kharif crops** (Paddy/Rice, Maize, Jowar, Bajra, Soybean, Groundnut, Cotton, Arhar/Pigeon pea).
4. **Sharad (Autumn — शरद)**: - *Period*: Ashwina – Kartika (approx. mid-August to mid-October). - *Climate*: Withdrawal of monsoon clouds; clear deep blue skies; high daytime temperatures accompanied by oppressive humidity (**"October Heat"**); cooling night breezes. - *Nature*: Blooming of white water lilies (*Kash* flowers); calm waters; celebration of Sharad Purnima and Durga Puja. - *Agriculture*: Grain filling and harvesting of early Kharif crops.
5. **Hemanta (Pre-Winter / Late Autumn — हेमन्त)**: - *Period*: Margashirsha – Pausha (approx. mid-October to mid-December). - *Climate*: Gradual decline in diurnal temperatures, pleasant sunny afternoons, briskly cold nights, morning dew condensation on grass. - *Nature*: Deciduous leaves begin turning gold/brown; wild animals store body fat; arrival of early migratory water birds. - *Agriculture*: Deep field ploughing and sowing of primary **Rabi crops** (Wheat, Gram, Mustard, Peas, Linseed, Potato).
6. **Shishira (Winter / Cold Season — शिशिर)**: - *Period*: Magha – Phalguna (approx. mid-December to mid-February). - *Climate*: Peak winter chill; intense ground frost, freezing temperatures in northern foothills, dense morning radiation fog, chilling north-westerly winds. - *Atmosphere*: Western Disturbances origin from the Mediterranean Sea bring vital winter rain showers (**Mahawat**), which are immensely beneficial for the vegetative growth of wheat. - *Nature*: Trees stand bare; dense concentrations of migratory waterfowl (Siberian cranes, Bar-headed geese, Pintails) flocking to wetlands like Keoladeo Ghana (Bharatpur) and Sur Sarovar (Agra). - *Agriculture*: Vegetative tillering of wheat; harvest of sugarcane and winter vegetables (carrots, radishes, cauliflower).

18. Comprehensive Directory of National & International Science, Agricultural & Medical Institutes

Institute Name	Hindi / Acronym	Exact Location	Administrative Jurisdiction & High-Frequency Must-Score Focus
Indian Veterinary Research Institute	IVRI / भारतीय पशु चिकित्सा अनुसंधान संस्थान	**Izatnagar (Bareilly), UP**	Premier national veterinary medicine & animal health research institute under ICAR.
Central Avian Research Institute	CARI / केंद्रीय पक्षी अनुसंधान संस्थान	**Izatnagar (Bareilly), UP**	Advanced poultry science and avian research under ICAR.
Central Drug Research Institute	CDRI / केंद्रीय औषधि अनुसंधान संस्थान	**Lucknow, UP**	CSIR laboratory; developed **Centchroman (Saheli)**, world's first non-steroidal oral contraceptive pill.
Central Institute of Medicinal and Aromatic Plants	CIMAP / केंद्रीय औषधीय एवं सुगंध पौधा संस्थान	**Lucknow, UP**	CSIR institute focusing on bioprospection and processing of medicinal/aromatic flora (established 1957 as CIMPO).
National Botanical Research Institute	NBRI / राष्ट्रीय वनस्पति अनुसंधान संस्थान	**Lucknow, UP**	Premier CSIR research laboratory for plant taxonomy, modern botanical biology, and floriculture.

Institute Name	Hindi / Acronym	Exact Location	Administrative Jurisdiction & High-Frequency Must-Score Focus
Birbal Sahni Institute of Palaeosciences	BSIP / बीरबल साहनी पुराविज्ञान संस्थान	**Lucknow, UP**	Founded by renowned Indian paleobotanist **Prof. Birbal Sahni**; autonomous research institute under DST (non-CSIR).
Indian Institute of Sugarcane Research	IISR / भारतीय गन्ना अनुसंधान संस्थान	**Lucknow, UP**	ICAR apex institute for fundamental and applied sugarcane agronomy.
Central Institute of Subtropical Horticulture	CISH / केंद्रीय उपोष्ण बागवानी संस्थान	**Lucknow, UP**	ICAR institute (originated 1972 as Central Mango Research Station) for mango, guava, and subtropical fruits.
National Sugar Institute	NSI / राष्ट्रीय शर्करा संस्थान	**Kanpur, UP**	Premier institute under Ministry of Consumer Affairs, Food & Public Distribution for sugar engineering & alcohol tech.
Indian Institute of Pulses Research	IIPR / भारतीय दलहन अनुसंधान संस्थान	**Kanpur, UP**	ICAR apex institute for chickpea, pigeonpea, lentil, and pulses crop breeding and improvement.
Artificial Limbs Manufacturing Corporation of India	ALIMCO / भारतीय कृत्रिम अंग निर्माण निगम	**Kanpur, UP**	Founded in 1972 under Ministry of Social Justice & Empowerment; manufactures Jaipur foot, prosthetics, and orthotic aids.
Indian Institute of Vegetable Research	IIVR / भारतीय सब्जी अनुसंधान संस्थान	**Varanasi, UP**	ICAR national research institute for vegetable breeding and crop management.
Central Institute for Research on Goats	CIRG / केंद्रीय बकरी अनुसंधान संस्थान	**Makhdoom, Mathura, UP**	ICAR institute established in 1979 for goat husbandry, reproduction, and caprine nutrition.
National JALMA Institute for Leprosy & Other Mycobacterial Diseases	Central Leprosy Institute	**Agra, UP**	Apex national biomedical institute for leprosy (*Mycobacterium leprae*) and tuberculosis diagnostics under ICMR.
National Bureau of Plant Genetic Resources	NBPGR	**New Delhi**	ICAR nodal institute for plant germplasm exploration, collection, quarantine, and long-term genebank storage.
National Institute of Immunology	NII	**New Delhi**	Autonomous institute under DBT; vaccine development, immunology, and host defense mechanisms.
National Institute of Plant Genome Research	NIPGR	**New Delhi**	Apex DBT research institute in computational and functional plant genomics.
National Brain Research Centre	NBRC	**Manesar (Gurugram), Haryana**	Dedicated neuroscience and brain mapping institute under DBT.
National Dairy Research Institute	NDRI	**Karnal, Haryana**	ICAR premier dairy institute; achieved India's first buffalo clones (*Samrupa, Garima*) and Gir cow clone (*Ganga*).
National Bureau of Animal Genetic Resources	NBAGR	**Karnal, Haryana**	Apex ICAR institution for genetic characterization and conservation of indigenous livestock and poultry breeds.
National Institute of Virology	NIV	**Pune, Maharashtra**	ICMR apex virology research laboratory; WHO H5 reference laboratory for South-East Asia.
National Centre for Cell Science	NCCS	**Pune, Maharashtra**	Premier DBT cell repository, cell biology, and tissue culture research facility.
Centre for DNA Fingerprinting and Diagnostics	CDFD	**Hyderabad, Telangana**	DBT autonomous biotechnology center founded by **Dr. Lalji Singh** for forensic DNA typing and molecular diagnostics.
Centre for Cellular and Molecular Biology	CCMB	**Hyderabad, Telangana**	Flagship CSIR institute for molecular genetics, proteomics, and modern biological sciences.

Institute Name	Hindi / Acronym	Exact Location	Administrative Jurisdiction & High-Frequency Must-Score Focus
National Institute for the Empowerment of Persons with Intellectual Disabilities	NIEPID (Formerly NIMH)	**Secunderabad, Telangana**	National institute for intellectual disability rehabilitation under Ministry of Social Justice & Empowerment.
National Centre for Biological Sciences	NCBS	**Bengaluru, Karnataka**	Premier biological sciences research center of Tata Institute of Fundamental Research (TIFR).
Institute of Life Sciences	ILS	**Bhubaneswar, Odisha**	Autonomous biotechnology research institute under DBT.
Institute of Microbial Technology	IMTECH	**Chandigarh**	CSIR center dedicated to industrial fermentation, microbial genomics, and protein engineering.
Rajiv Gandhi Centre for Biotechnology	RGCB	**Thiruvananthapuram, Kerala**	Premier national biotechnology institute under DBT focused on molecular medicine and disease biology.
High Security Animal Disease Laboratory	NIHSAD	**Bhopal, Madhya Pradesh**	Biosafety Level-4 (BSL-4) veterinary facility under ICAR for avian influenza and exotic viral animal pathogens.
Survey of India Headquarters	भारतीय सर्वेक्षण विभाग	**Dehradun, Uttarakhand**	India's National Mapping Agency, founded in 1767; operates under Ministry of Science and Technology.
National Institute for the Empowerment of Persons with Visual Disabilities	NIEPVD	**Dehradun, Uttarakhand**	Apex institute under Ministry of Social Justice & Empowerment for visually impaired rehabilitation and Braille printing.
Central Potato Research Institute	CPRI / केंद्रीय आलू अनुसंधान संस्थान	**Shimla, Himachal Pradesh**	ICAR institute for potato breeding, seed production, and disease surveillance.
National Rice Research Institute	NRRI / राष्ट्रीय चावल अनुसंधान संस्थान	**Cuttack, Odisha**	Apex ICAR institute for rice genetics, agronomy, and crop varieties (formerly CRRI).
Central Salt and Marine Chemicals Research Institute	CSMCRI	**Bhavnagar, Gujarat**	CSIR laboratory for marine chemicals, salt extraction, and desalination technologies.
Central Electrochemical Research Institute	CECRI	**Karaikudi, Tamil Nadu**	CSIR national laboratory dedicated to electrochemistry, batteries, fuel cells, and corrosion prevention.
Sugarcane Breeding Institute	SBI	**Coimbatore, Tamil Nadu**	Pioneer ICAR institute (founded 1912) for developing commercial hybrid sugarcane cultivars (*Co* series).
Institute of Forest Genetics and Tree Breeding	IFGTB	**Coimbatore, Tamil Nadu**	Apex national forestry institute under ICFRE.
International Rice Research Institute	IRRI & International Rice Genebank	**Los Baños, Philippines**	Developed Super Rice and Semi-dwarf high-yielding rice; houses the world's largest rice germplasm bank.
International Potato Center	CIP (Centro Internacional de la Papa)	**Lima, Peru**	CGIAR international research center dedicated to potato, sweetpotato, and Andean root crops.
Food and Agriculture Organization	FAO Headquarters	**Rome, Italy**	United Nations specialized agency leading international efforts to defeat hunger and ensure food security.
World Health Organization	WHO Headquarters	**Geneva, Switzerland**	United Nations specialized agency directing international public health.

19. Medical Diagnostic Devices, Clinical Instrumentation & Biomedical Technologies

Diagnostic Tool / Instrument	Operational Physics & Physiological Principle	Primary Clinical Application & Diagnostic Utility	High-Yield UP-PCS Must-Score
Stethoscope	Acoustic transmission and amplification of sound via mechanical chestpiece (diaphragm/bell).	Listening to internal physiological sounds: heartbeats (valvular sounds S_1, S_2), lung breath sounds, bowel motility.	Invented by **René Laennec** in 1816; operates on the principle of **multiple reflection of sound waves**.
Sphygmomanometer	Inflatable occluding cuff measuring counter-pressure against arterial pulsation (auscultation of Korotkoff sounds).	Quantitative measurement of arterial blood pressure ($mmHg$); normal resting adult value: $120/80\ \text{mmHg}$.	Contains **Mercury (Hg)**; measures systolic (ventricular contraction) over diastolic (ventricular relaxation) pressure.
Computed Tomography (CT Scan)	Computer-processed tomographic cross-sectional image synthesis using penetrating **X-ray beams**.	High-resolution cross-sectional visualization of brain, thorax, abdomen, bone trauma, tumors, and hemorrhage.	Uses **ionizing X-radiation**; developed by **Godfrey Hounsfield and Allan Cormack** (Nobel Prize in 1979).
Magnetic Resonance Imaging (MRI)	Uses powerful superconducting **magnetic fields** and **radiofrequency (RF) pulses** to align and detect hydrogen nuclei (protons) precession.	Extremely high contrast resolution imaging of soft tissues, brain parenchyma, spinal cord, joints, ligaments, and deep tumors.	**NO ionizing radiation** (completely free of X-rays or Gamma rays); based on the physical phenomenon of **Nuclear Magnetic Resonance (NMR)**.
Electrocardiogram (ECG / EKG)	Surface electrodes detect extracellular ionic current flows generated during myocardial depolarization and repolarization.	Detecting cardiac arrhythmias, myocardial infarction (heart attack), bundle branch blocks, and ischemia.	Developed by **Willem Einthoven** (Nobel Prize in 1924); primary wave deflections: **P wave** (atrial depolarization), **QRS complex** (ventricular depolarization), **T wave** (ventricular repolarization).
Electroencephalogram (EEG)	Scalp electrodes record spontaneous fluctuating electrical potentials generated by postsynaptic potentials of cerebral cortical neurons.	Diagnosing epilepsy, focal seizures, sleep disorders, encephalopathies, and confirming clinical **brain death**.	Pioneered by German psychiatrist **Hans Berger** in 1924; records alpha (α), beta (β), theta (θ), and delta (δ) brain waves.
Electrooculography (EOG)	Electrodes placed around the orbit measure the standing steady electrical potential between the cornea (+ve) and retina (-ve).	Measuring horizontal/vertical eye movements, nystagmus, and assessing electrical function of the retinal pigment epithelium.	The resulting diagnostic recording is termed the **electrooculogram**.
Electromyography (EMG)	Needle or surface electrodes evaluate electrical potentials generated by skeletal muscle fibers during rest and voluntary contraction.	Diagnosing neuromuscular diseases, peripheral neuropathies, myasthenia gravis, and amyotrophic lateral sclerosis (ALS).	Record generated by an electromyograph is called an **electromyogram**.
Audiogram	Pure-tone audiometer measures auditory sensory threshold intensity across standardized frequencies.	Evaluating hearing acuity, distinguishing conductive from sensorineural hearing loss.	**X-axis** = Frequency in Hertz (Hz); **Y-axis** = Intensity in Decibels (dB) relative to normal hearing curve.
Mammography	Low-dose, low-energy **X-ray examination** with breast tissue compression.	Early screening and detection of subclinical breast cancer, microcalcifications, and architectural breast distortion.	Recommended routinely for early detection of female malignancies; the image is termed a **mammogram**.
Endoscope	Slender, flexible tubular probe utilizing bundles of optical glass fibres to transmit light and return images to an external monitor.	Non-surgical direct visualization and biopsy of internal luminal tracts (digestive tract: gastroscopy, colonoscopy; respiratory: bronchoscopy).	Operates strictly on the optical principle of **Total Internal Reflection (TIR)**; cannot be categorized merely as a standard camera or telescope.

Diagnostic Tool / Instrument	Operational Physics & Physiological Principle	Primary Clinical Application & Diagnostic Utility	High-Yield UP-PCS Must-Score
Cardiograph	Mechanical or electronic transducer registering heart motion curves.	Graphically registers mechanical movements and volume changes of the contracting heart.	Instrument that draws cardiac displacement curves.
Crescograph	Mechanical/optical magnification mechanism to record minute macroscopic plant growth increments.	Measuring precise instantaneous elongation and growth responses in living plants under toxic or stimulatory agents.	Invented in the early 20th century by legendary Indian scientist **Sir Jagdish Chandra Bose (J.C. Bose)**.
Bone Mineral Density (BMD) Test	Dual-energy X-ray Absorptiometry (DEXA) measuring bone mineral mass per square centimeter.	Diagnosing **Osteoporosis** (brittle bone disease marked by microstructure deterioration and low bone density) and assessing fracture risk.	High-frequency PSC repeat: BMD is exclusively performed for **Osteoporosis**, evaluating bone calcium/mineral concentration.
Lithotripsy (ESWL)	Focused high-energy acoustic **extracorporeal shockwaves (ultrasound shockwaves)** pulverize solid mineral deposits into sand-sized fragments.	Non-invasive disintegration of renal calculi (**kidney stones**), ureteral stones, and bladder stones, passing fragments painlessly through urine.	From Greek *lithos* = stone, *tripsis* = crushing; eliminates need for surgical open nephrolithotomy.
Electron Beam Therapy (EBT)	Linear accelerator emits high-energy beams of electrons directed externally to superficial tissue targets.	Radiation therapy used to treat superficial cancers such as **cutaneous T-cell lymphoma (mycosis fungoides)** and skin carcinomas without deep tissue toxicity.	Electrons have a finite, steep depth dose fall-off, shielding underlying visceral organs from radiation damage.
Barium Meal X-Ray	Patient ingests radio-opaque **Barium Sulfate ($BaSO_4$)** suspension before gastrointestinal X-ray fluoroscopy.	Diagnosing ulcers, esophageal strictures, hiatal hernias, and gastrointestinal tumors.	Barium has a high atomic number ($Z = 56$); acts as an **excellent absorber of X-rays**, creating sharp white radiographic contrast.

20. Important Observance Days in Health, Science & Environment

Date	International / National Observance Day	Significance, Landmark Theme & UP-PCS Focus
Last Sunday of January	**World Leprosy Day** (Anti-Leprosy Day in India observed on **30 January**)	Observed on the martyrdom anniversary of Mahatma Gandhi in India; raising awareness on *Mycobacterium leprae* elimination.
4 February	**World Cancer Day**	Led by the Union for International Cancer Control (UICC); promoting early oncological screening, prevention, and equitable access.
8 March	**International Women's Day**	UN observance highlighting women's rights, socio-economic empowerment, and maternal/adolescent healthcare.
22 March	**World Water Day**	Focuses on the importance of freshwater, sustainable aquifer management, and global water sanitation.
24 March	**World Tuberculosis (TB) Day**	Commemorates **Dr. Robert Koch's discovery of *Mycobacterium tuberculosis*** on 24 March 1882; DOTS strategy focus.
7 April	**World Health Day**	Marks the founding anniversary of the **World Health Organization (WHO)** on 7 April 1948.
10 April	**World Homeopathy Day**	Commemorates the birth anniversary of **Dr. Christian Friedrich Samuel Hahnemann**, the founder of Homeopathy.
22 April	**World Earth Day**	Global environmental awareness day initiated in 1970; focuses on environmental protection and climate action.
25 April	**World Malaria Day**	Commemorates the global fight against *Plasmodium* transmission by female *Anopheles* mosquitoes.

Date	International / National Observance Day	Significance, Landmark Theme & UP-PCS Focus
7 May	**World Athletics Day**	Promotes physical fitness and sports participation among school and collegiate youth.
8 May	**World Red Cross Day**	Birth anniversary of **Henry Dunant**, founder of the International Committee of the Red Cross (ICRC).
8 May	**World Thalassemia Day**	Dedicated to raising awareness on inherited hemoglobinopathies and genetic counseling.
5 June	**World Environment Day**	Established by the UN General Assembly at the 1972 Stockholm Conference on the Human Environment; landmark ecology event.
21 June	**International Day of Yoga**	UN-declared day initiated by India in 2014; coincides with the summer solstice in the Northern Hemisphere.
16 July	**ICAR Foundation & Technology Day**	Commemorates the founding of the **Indian Council of Agricultural Research** on 16 July 1929.
16 September	**International Ozone Day**	Commemorates the signing of the **Montreal Protocol on Substances that Deplete the Ozone Layer** (16 September 1987).
21 September	**World Alzheimer's Day**	Raises global awareness on neurodegenerative dementia and Alzheimer's disease support.
4 October	**World Animal Day**	Commemorates the feast day of Francis of Assisi, patron saint of animals; focuses on animal welfare and wildlife conservation.
10 October	**World Mental Health Day**	Initiated by the World Federation for Mental Health to combat stigma and foster psychiatric care.
12 October	**World Arthritis Day**	Raising awareness on musculoskeletal degenerative disorders, osteoarthritis, and rheumatoid arthritis.
16 October	**World Food Day**	Marks the founding date of the UN **Food and Agriculture Organization (FAO)** in 1945; combating malnutrition and food insecurity.
24 October	**World Polio Day**	Commemorates the birth of **Jonas Salk**, who developed the first inactivated polio vaccine (IPV).
14 November	**World Diabetes Day**	Commemorates the birthday of **Sir Frederick Banting**, co-discoverer of insulin with Charles Best in 1921.
1 December	**World AIDS Day**	Global campaign dedicated to raising awareness of AIDS caused by HIV infection; Red Ribbon symbol.
18 December	**World Minority Rights Day**	Commemorates the 1992 UN Declaration on the Rights of Persons Belonging to National or Ethnic, Religious and Linguistic Minorities.

21. Major Agricultural & Production Revolutions in India

Agricultural / Industrial Revolution	Commodity / Product Field	Key Pioneers / Leadership & Core Strategy
Green Revolution (हरित क्रांति)	**Cereal Grains (Wheat & Rice)**	**Dr. M.S. Swaminathan** (India), **Dr. Norman Borlaug** (Global); Semi-dwarf HYV seeds, chemical fertilizers, tubewell irrigation.
White Revolution (श्वेत क्रांति / Operation Flood)	**Milk and Dairy Production**	**Dr. Verghese Kurien** ("Father of White Revolution"); National Dairy Development Board (NDDB), Anand cooperative model.
Yellow Revolution (पीली क्रांति)	**Oilseeds Production** (Mustard, Sunflower, Soybean)	**Sam Pitroda** (Technology Mission on Oilseeds, 1986); aimed at achieving self-sufficiency in edible oils.
Blue Revolution (नीली क्रांति)	**Fisheries and Aquaculture**	**Dr. Hiralal Chaudhari** and **Dr. Arun Krishnan**; intensive inland pisciculture and coastal marine fisheries.
Pink Revolution (गुलाबी क्रांति)	**Meat, Poultry Processing / Onions / Prawns**	**Durgesh Patel**; modernization of meat hygiene, refrigerated transport, and brackishwater prawn farming.

Agricultural / Industrial Revolution	Commodity / Product Field	Key Pioneers / Leadership & Core Strategy
Black Revolution (काली क्रांति)	**Petroleum & Crude Oil Production**	Promoting bio-ethanol blending with petrol (20% ethanol blending target) to cut fossil fuel imports.
Silver Revolution (रजत क्रांति)	**Egg and Poultry Production**	**Indira Gandhi**; genetic breeding of high-laying poultry strains and veterinary feed security.
Red Revolution (लाल क्रांति)	**Meat and Tomato Production**	**Vishal Tewari**; processing modernization and post-harvest cold chain logistics for tomatoes and livestock.
Round Revolution (गोल क्रांति)	**Potato Production**	CPRI Shimla hybrid breeding; boosting production of tubers across northern alluvial belts.
Grey Revolution (धूसर क्रांति)	**Chemical Fertilizers Production**	Expanding domestic production of NPK chemical fertilizers to sustain high-yield cereal crops.
Brown Revolution (भूरी क्रांति)	**Leather Industry / Non-Conventional Energy / Cocoa**	Eco-friendly processing of hides, rural tanneries, and sustainable tribal agro-forestry.
Golden Revolution (स्वर्ण क्रांति)	**Horticulture, Honey (Apiculture), & Fruits**	**Nirpakh Tutej**; integrated fruit orchards, floriculture, high-value vegetable farming, and bee-keeping.
Golden Fibre Revolution (स्वर्ण रेशा क्रांति)	**Jute Production**	Modernizing cultivation of *Corchorus* in the Bengal delta; eco-friendly packaging and biodegradable sacks.
Silver Fibre Revolution (रजत रेशा क्रांति)	**Cotton Production**	Adoption of Bt cotton hybrids, integrated pest management, and expanding long-staple cotton spinning.

22. Applied Animal Husbandry, Veterinary Science & Wildlife Conservation

- **Bovine Artificial Insemination & Cryopreservation**:
- Elite bull semen for artificial insemination is stored cryogenically in **Liquid Nitrogen at $-196^{\circ}\mathrm{C}$ ($-320^{\circ}\mathrm{F}$)** inside vacuum-insulated containers.
- *Must-Score*: Cryopreservation in liquid nitrogen preserves semen viability and motility indefinitely with **zero fertility deterioration over decades**; storage in dry ice/alcohol ($-100^{\circ}\mathrm{F}$) leads to gradual fertility loss.
- **Indigenous Cattle & Breed Identification**:
- **Rashtriya Gokul Mission**: Launched by the Union Government under the National Programme for Bovine Breeding and Dairy Development; dedicated to **conserving, improving, and propagating indigenous cow breeds** (e.g. Gir, Sahiwal, Red Sindhi, Tharparkar) in a scientific manner.
- **Gir Breed Cloned Cow 'Ganga'**: Born on 16 March 2023 at NDRI Karnal, India's first cloned calf of the indigenous Gir breed, capable of producing > 15 litres of milk per day.
- **Highest Milk-Yielding Dairy Cow**: **Holstein-Friesian** (native to Netherlands and northern Germany), yielding an average of over $11,000$ kg of milk per lactation cycle.
- **Breed Classification Fact**:
  - **Cow Breeds**: Gir, Sahiwal, Red Sindhi, Tharparkar, Nagauri, Kankrej, Rathi.
  - **Buffalo Breeds**: Murrah, Bhadawari, Jaffarabadi, Surti, Mehsana, Nili-Ravi.
  - **Goat Breeds**: Jamunapari, Barbari, Beetal, Black Bengal.
  - **Sheep Breeds**: Dorset, Merino, Marwari, Deccani, Lohi.
- **Kadaknath Chicken (Kali Masi)**:
- Rare indigenous poultry breed native to the tribal districts of **Jhabua and Dhar, Madhya Pradesh**.
- Characterized by **melanism (black plumage, skin, beak, bones, and internal organs/meat)** due to excessive melanin deposition. Granted a prestigious **Geographical Indication (GI Tag)** by the Government of India in 2018. High in protein ($> 25\%$) and iron, low in cholesterol ($0.73 - 1.05\%$).
- **Camel Adaptations & International Year of Camelids**:
- **Hump Function**: The camel's prominent dorsal hump (up to 50 kg) stores **pure adipose fat**, NOT water. Camels metabolize this fat tissue into metabolic water and energy during extended desert journeys.
- **World's First Camel Hospital**: Founded in **Dubai, United Arab Emirates (2017)**, featuring state-of-the-art specialized veterinary surgical suites for dromedaries.

- **International Year of Camelids (2024)**: Designated by the **United Nations** to recognize the indispensable role of camelids (dromedaries, Bactrian camels, llamas, alpacas) in supporting nomadic pastoral livelihoods in arid and mountainous zones.
- **Wildlife Conservation & Extinction Milestones**:
- **Project Cheetah**: Asiatic cheetah (*Acinonyx jubatus venaticus*) was officially declared extinct in India in **1952** due to excessive sport-hunting and habitat destruction. In September 2022, the Government of India reintroduced African cheetahs translocated from **Namibia, South Africa, and Botswana** into **Kuno National Park**, Madhya Pradesh.
- **Sixth Mass Extinction (Anthropocene Extinction)**: The ongoing catastrophic die-off of terrestrial and marine biodiversity driven almost entirely by **anthropogenic activities**: habitat destruction, overexploitation/poaching, pollution, monoculture agriculture, invasive alien species, and global climate change.
- **Asian Giant Tortoise (*Manouria emys*)**: Mainland Asia's largest tortoise species; captive-bred individuals were re-wilded into **Ntangki National Park**, Nagaland (December 2022).
- **State-Level Committee for Vulture Conservation (SLCVC)**: Established by **Tamil Nadu** (2022) to enforce the National Action Plan for Vulture Conservation (APVC 2020–2025) and curb veterinary use of toxic NSAIDs like diclofenac, aceclofenac, and ketoprofen.

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23. Ethology, Insect Communication & Environmental Biology

- **Honeybee Communication & Karl von Frisch's Discoveries**:
- Honeybees (*Apis* genus) are eusocial hymenopteran insects organized into distinct castes: 1 fertile Queen, hundreds of fertile male Drones, and tens of thousands of sterile female Worker bees.
- **Colony Recognition**: Members of a beehive recognize fellow nestmates primarily by a **colony-specific pheromonal odor (chemical smell)**, preventing unauthorized intrusions by foragers from foreign hives.
- **Deciphering the Dance Language (Nobel Prize 1973)**: Austrian-German ethologist **Karl von Frisch** proved that foraging worker bees communicate the location, distance, and direction of newly discovered nectar and pollen sources to hivemates through specialized choreographic body movements:
  1. **Round Dance (Kreistanz)**: The bee executes rapid small circles on the comb, alternating right and left. Indicates a rich food source located **close to the hive (within approximately 75 meters)** without encoding directional azimuth.
  2. **Waggle Dance (Schwänzeltanz)**: The bee runs straight forward across a figure-eight path while vigorously waggling her abdomen from side to side, then returns in a loop. Communicates two vital parameters: - **Direction**: The angle of the straight waggle run relative to vertical gravity matches the exact angle of the flower patch relative to the sun's horizontal azimuth. - **Distance**: The duration and frequency of the waggle run directly correlate with the distance to the distant foraging ground.
- **Landmine Detection**: Honeybees possess an extraordinary olfactory sensitivity and can be trained via Pavlovian reward conditioning to detect airborne vapors of 2,4-dinitrotoluene (DNT) leaking from buried landmines.
- **Flight Speed**: Average foraging flight speed is approximately **12–15 miles/hour (17–21 km/h)**.
- **Biochemical Trivia & Applied Pharmacology**:
- **Saheli (Centchroman / Ormeloxifene)**: World's first indigenously synthesized, non-steroidal, non-hormonal, once-a-week oral contraceptive pill developed by the **Central Drug Research Institute (CDRI), Lucknow** (introduced commercially in 1991).
- **Common Contraceptive Formulations**: Modern oral and emergency contraceptive pills predominantly employ the synthetic progestogen **Levonorgestrel**.
- **Sulphur Mustard (Mustard Gas)**: Premier blistering chemical warfare agent deployed by the German army in 1917 during World War I; volatile oily liquid, potent vesicant (causes severe chemical skin burns and deep lung ulcerations within 24 hours), highly mutagenic, lipophilic, and carcinogenic.
- **Shanti Swarup Bhatnagar Prize**: Founded in 1958 by CSIR in honor of its founder-director **Dr. Shanti Swarup Bhatnagar**; India's most prestigious annual award for outstanding multidisciplinary scientific achievements in Biology, Chemistry, Physics, Medicine, Mathematics, Engineering, and Earth Sciences (awarded to scientists under 45 years of age).
- **Eye Donation / Corneal Transplantation**: In human eye donation, only the **Cornea** (the avascular, clear, curved transparent window forming the anterior sixth of the fibrous tunic) is excised and grafted (**Keratoplasty**); does not provoke immunological tissue rejection due to its complete avascularity.
- **Clinical Dehydration**: Acute excessive loss of body fluids and essential electrolytes, predominantly **Sodium Chloride ($\mathbf{NaCl}$)**. A severely dehydrated person must never consume hypertonic seawater (which causes rapid cellular plasmolysis and accelerated fatal dehydration) or excessive diuretic stimulants like caffeine/alcohol.
- **Structural Keratins**: Hair, nails, hooves, animal horns, claws, feathers, and spider silk are composed of durable fibrous **Keratin proteins**; cotton, by contrast, is pure vegetal **cellulose** (polysaccharide).
- **Rayon (Artificial Silk)**: Manufactured fiber regenerated from natural plant cellulose (wood pulp); it is **NOT modified starch**.

24. Nobel Prize in Physiology or Medicine: Landmarks & Historical Frontiers

- **Nobel Institution & Ceremony**: Awarded annually by the **Nobel Assembly at the Karolinska Institute**, Stockholm, Sweden, on **10 December** (the death anniversary of Alfred Nobel).
- **First Nobel Prize in Medicine (1901)**: Awarded to German physiologist **Emil von Behring** for his discovery of **serum therapy** and developing an antitoxin against diphtheria (*Corynebacterium diphtheriae*).
- **First Female Laureate in Medicine (1947)**: **Gerty Theresa Cori** (along with Carl Cori and Bernardo Houssay) for discovering the enzymatic pathway of glycogen metabolism (**the Cori Cycle**).
- **Pioneering Genetic & Ethological Discoveries**:
- **1959**: **Arthur Kornberg** (with Severo Ochoa) for the *in vitro* enzymatic synthesis of DNA using DNA polymerase.
- **1962**: **James Watson, Francis Crick, and Maurice Wilkins** for resolving the double-helix molecular model of DNA.
- **1968**: **Har Gobind Khorana** (with Marshall Nirenberg and Robert Holley) for deciphering the genetic code and its role in cellular protein synthesis.
- **1973**: **Karl von Frisch, Konrad Lorenz, and Nikolaas Tinbergen** for discoveries concerning the organization and elicitation of individual and social behavior patterns (waggle dance of honeybees).
- **1978**: **Werner Arber, Daniel Nathans, and Hamilton O. Smith** for discovering restriction endonucleases (molecular scissors).
- **1984**: **Georges Köhler and César Milstein** (with Niels Jerne) for inventing **Hybridoma Technology** for the mass production of monoclonal antibodies.
- **Recent Groundbreaking Nobel Frontiers**:
- **2024 Nobel Prize**: Awarded jointly to **Victor Ambros and Gary Ruvkun** for the landmark discovery of **microRNA (miRNA)** and its fundamental role in post-transcriptional gene regulation.
- **2025 Nobel Prize**: Awarded jointly to **Mary E. Brunkow, Frederick J. Ramsdell, and Shimon Sakaguchi** for their groundbreaking discoveries concerning **peripheral immune tolerance**, specifically identifying **Regulatory T cells (Tregs)** and the master regulator gene **FOXP3**, which act as "security guard" cells preventing autoimmune attacks against the body's own tissues.

Bilingual Terminology -- Microbiology, Environment and Applied Science

English Term	Hindi Term	Key Anchor
Bacteria (जीवाणु)	जीवाणु / बैक्टीरिया	Prokaryotic unicellular; most ancient; both helpful and harmful
Virus (विषाणु)	विषाणु	Acellular; DNA or RNA + protein coat; obligate intracellular parasite
Fungi (कवक)	कवक	Eukaryotic; heterotrophic; chitin cell wall; includes moulds, yeasts
Protozoa (प्रोटोजोआ)	एककोशिकीय प्राणी	Unicellular eukaryotes; some are pathogens (Plasmodium, Entamoeba)
Algae (शैवाल)	शैवाल	Photosynthetic; mostly aquatic; no true roots/stems/leaves
Fermentation (किण्वन)	किण्वन	Anaerobic breakdown of sugars by yeast/bacteria; produces alcohol/acid
Pasteurisation (पास्चुरीकरण)	पास्चुरीकरण	Heating food to kill pathogens without destroying nutritional value; Pasteur
Sterilisation (निर्जर्मीकरण)	निर्जर्मीकरण	Complete destruction of all microorganisms including spores
Antibiotic (प्रतिजैविक)	प्रतिजैविक	Kills/inhibits bacteria; from microorganisms (fungi, bacteria)
Vaccine (टीका)	टीका	Immunisation preparation; stimulates immune memory
Biodegradable (जैव-अपघटनीय)	जैव-अपघटनीय	Can be broken down by microorganisms; e.g., paper, food waste
Non-Biodegradable (अजैव-अपघटनीय)	अजैव-अपघटनीय	Cannot be broken down biologically; e.g., plastics, glass, DDT
Bioremediation (जैव उपचारण)	जैव उपचारण	Using microorganisms to clean up pollutants
Nitrogen Cycle (नाइट्रोजन चक्र)	नाइट्रोजन चक्र	Cycling of nitrogen through atmosphere, soil, plants, animals
Carbon Cycle (कार्बन चक्र)	कार्बन चक्र	Cycling of carbon through photosynthesis, respiration, decomposition
Food Web (आहार जाल)	आहार जाल	Complex network of food chains in an ecosystem
Biogas (बायोगैस)	बायोगैस	Methane (55-75%) + CO2; produced by anaerobic digestion of organic matter

English Term	Hindi Term	Key Anchor
Composting (खाद बनाना)	कम्पोस्टिंग	Aerobic decomposition of organic matter to make compost/manure
Ecosystem (पारितंत्र)	पारितंत्र	Community of living organisms + their physical environment
Biodiversity (जैव विविधता)	जैव विविधता	Variety of life on Earth at genetic, species, and ecosystem levels

One-Liner Revision -- Microbiology, Environment and Applied Science

#	Fast Fact
1	Bacteria discovered by = **Antonie van Leeuwenhoek** (1676, using microscope)
2	Germ Theory of Disease = **Louis Pasteur** (France) and Robert Koch (Germany)
3	Koch's Postulates = rules to establish pathogen causes a specific disease
4	Penicillin = from *Penicillium notatum*; discovered by **Alexander Fleming (1928)**
5	Streptomycin = discovered by **Selman Waksman (1943)**; Nobel 1952; treats TB
6	Pasteurisation (milk) = 72°C for 15 seconds (HTST) OR 63°C for 30 min (LTST)
7	Virus = DNA OR RNA (never both) + protein coat (capsid); no metabolism outside host
8	Largest virus = Mimivirus / Pandoravirus; Smallest virus = Parvovirus
9	Prions = misfolded proteins; cause Creutzfeldt-Jakob Disease (CJD) / mad cow disease
10	Viroids = smallest known infectious agents; only RNA, no protein coat; infect plants
11	Mycology = study of fungi; Bacteriology = study of bacteria; Virology = study of viruses
12	Yeast = unicellular fungus; used in baking (CO2) and brewing (ethanol)
13	Biogas composition = **Methane (55-75%)** + CO2 (25-45%) + trace gases
14	Biogas plants = use cow dung + water; anaerobic fermentation; produces CH4
15	Composting = aerobic breakdown; humus-rich dark material; improves soil fertility
16	Vermicomposting = earthworm-based composting; best compost quality
17	Nitrogen cycle steps: Fixation -> Nitrification -> Assimilation -> Ammonification -> Denitrification
18	Denitrification = conversion of nitrates back to N2 gas; by Pseudomonas denitrificans
19	Biodiversity hotspots globally = **36** (CI criteria: >1,500 endemic plants + >70% habitat lost)
20	India's biodiversity hotspots = **4**: Western Ghats, Eastern Himalayas, Indo-Burma, Sundaland
21	IUCN Red List categories: Extinct > EW > Critical > Endangered > Vulnerable > NT > LC
22	Project Tiger (1973) = conservation of Bengal Tiger; 53 tiger reserves in India
23	Convention on Biological Diversity (CBD) = **1992**, Rio de Janeiro; India ratified 1994
24	Ramsar Convention (1971) = protection of **wetlands**; Ramsar, Iran
25	India's Ramsar sites = **75+** (largest number of wetland sites in South Asia)
26	Biopesticide example: **Bacillus thuringiensis (Bt)** toxin against Lepidoptera larvae
27	Biofertiliser = Rhizobium, Azotobacter, Blue-Green Algae (BGA/Anabaena), Azolla
28	Eutrophication = excess N + P in water -> algae bloom -> hypoxia -> dead zone
29	BOD (Biochemical Oxygen Demand) = measures organic pollution in water; high BOD = more pollution
30	One-Health concept = human health, animal health, and environment are interconnected